

WEB INTEGRATED MOVIE RECOMMENDATION SYSTEM

An Industry Oriented Mini Project Report submitted to

Jawaharlal Nehru Technological University

in partial fulfillment of the requirements for the award of Degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

D.Vaishnavi (22WJ1A0562)

Under the Guidance of

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June, 2025

DEPARTMENT OF COMPUTER SCIENCE&ENGINEERING

CERTIFICATE

This is to certify that Industry Oriented Mini Project entitled “**WEB INTEGRATED MOVIE RECOMMENDATION SYSTEM**” is being presented with report by **D.Vaishnavi** bearing Roll.No. **22WJ1A0562** in partial fulfillment for the award of Degree of Bachelor of Technology in Computer Science & Engineering to the JawaharlalNehru Technological University is a record of Bonafide work carried out them under my guidance and supervision.

The results embodied in this project report have not been submitted to any other University or Institute for

the award of any Degree or Diploma.

Internal Guide

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PROJECT COMPLETION CERTIFICATE

This is to certify that the following students of final year B. Tech, Department Of **Computer Science and Engineering** – Guru Nanak Institutions Technical Campus have completed their training and project at GNITC successfully

STUDENT NAME

ROLL NO.

1. D.Vaishnavi

22WJ1A0562

The training was conducted on AI based Python Technology for the completion of the project titled in WEB INTEGRATED MOVIE RECOMMENDATION SYSTEM

In JUNE 2025 The project has been completed in all aspects.



Declaration of Student

I, D.Vaishnavi(22WJ1A0562), hereby declare that the major project titled “Web Integrated Movie Recommendation System” has been carried out by us as part of the requirements for the award of the Degree of B.Tech in the Department of Computer Science and Engineering at Guru Nanak Institutions Technical Campus.

I confirm the following:

1. The project was undertaken by us under the supervision of our guide, [Guide's Name], from the selection of the topic to the completion of the final report.
2. I have ensured that the results presented in the report are accurate and based on our original work.
3. Each member of the team has contributed significantly and appropriately to the project work.
4. The project report has been prepared with diligence, ensuring clarity, accuracy, and adherence to academic standards.

I further declare that this report has not been submitted, in part or full, to any other institution or university for the award of any degree or diploma.

D.Vaishnavi

Signature

22WJ1A0562

Date:

Place:

Declaration of Guide

I, Mr.LALU BANOTHU, hereby declare that I have guided the Industry Oriented Mini Project titled “Web Integrated Movie Recommendation System”undertaken by D.Vaishnavi (22WJ1A0562). This project was carried out towards the fulfillment of the requirements for the award of the Degree of B. Tech in Department of Computer Science and Engineering at Guru Nanak Institutions Technical Campus.

As the guide, I confirm the following:

1. I have overseen the entire project process, from the selection of the project title to the submission of the final report.
2. I have reviewed and certified the accuracy and relevance of the results presented in the report.
3. The contributions of each student have been appropriately recognized and assessed.
4. The project report has been prepared under my supervision, ensuring adherence to high standards of quality, clarity, and structure.

I further certify that this project report has not been previously submitted in part or full for the award of any degree or diploma by any institution or university.

Name of Guide: Mr.LALU BANOTHU

Signature of the Guide

Date:

Place:

ACKNOWLEDGEMENT

I would like to express our sincere gratitude to our internal guide, **Mr.LALU BANOTHU, Associate professor, Computer Science And Engineering**, for his valuable guidance, encouragement, and continuous support throughout the duration of this project.

We are very much grateful to our Project Coordinator, **Mr. Mohd Irfan, Assistant Professor of CSE, GNITC**, Hyderabad, who is not only shown utmost patience but was fertile in suggestions, vigilant in directions of error, and have been infinitely helpful.

We were also thankful to **Dr. Geeta Tripathi, Professor & HOD CSE, GNITC**, Hyderabad, for being so helpful with valuable insights and guidance during our project.

We wish to express our sincere thanks to **Dr. Rishi Sayal, Professor in CSE & Associate Director, GNITC** for providing us with the conducive environment for carrying through our academic schedules and project with ease.

On a more personal note, we thank our beloved parents and friends for their moral support during our project.

D.Vaishnavi

22WJ1A0562

ABSTRACT

The use of a Recommendation system is to provide users with recommendations based on their search preferences. In this article, I will introduce you to a machine learning project on the Netflix recommendation system with Python. Netflix is a company that manages a large collection of TV shows and movies, streaming it anytime via online. This business is profitable because users make a monthly payment to access the platform. However, customers can cancel their subscriptions at any time. Therefore, the company must keep the users hooked on the platform and not lose their interest. This is where recommendation systems start to play an important role, providing valuable suggestions to users is essential. Netflix's recommendation system helps them increase their popularity among service providers as they help increase number of items sold, offer a diverse selection of items, increase user satisfaction, as well as user loyalty to the company, and they are very helpful in getting a better understanding of what the user wants. Then it's easier to get the user to make better decisions from a wide variety of movie products. The idea behind the Netflix recommendation system is to recommend the most popular movies to users. They could be the most-watched, or also the ones with the highest ratings. The popularity of recommendations can be built based on usage data and article content.

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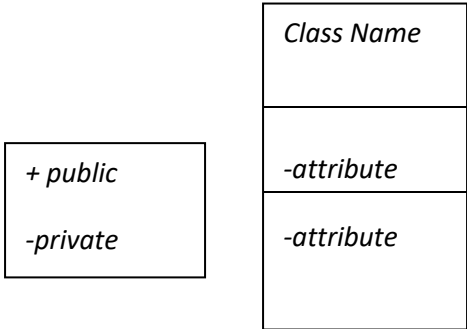
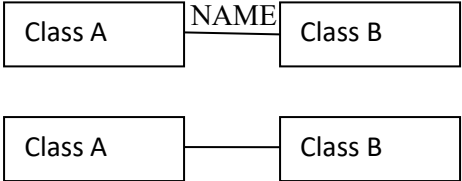
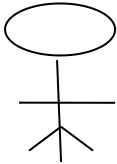
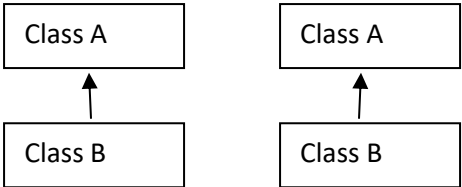
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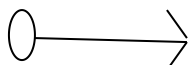
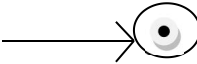
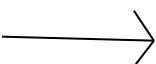
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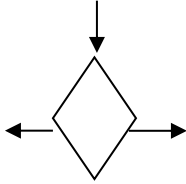
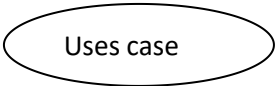
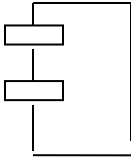
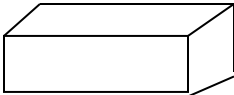
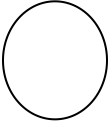
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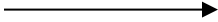
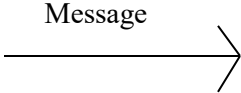
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LIST OF SYMBOLS

S.NO	NOTATION NAME	NOTATION	DESCRIPTION
1.	Class		Represents a collection of similar entities grouped together.
2.	Association		Associations represents static relationships between classes. Roles represents the way the two classes see each other.
3.	Actor		It aggregates several classes into a single classes.
4.	Aggregation		Interaction between the system and external environment

5.	Relation (uses)	uses	Used for additional process communication.
6.	Relation (extends)	extends 	Extends relationship is used when one use case is similar to another use case but does a bit more.
7.	Communication		Communication between various use cases.
8.	State		State of the processes.
9.	Initial State		Initial state of the object
10.	Final state		Final state of the object
11.	Control flow		Represents various control flow between the states.

12.	Decision box		Represents decision making process from a constraint
13.	Use case		Interact ion between the system and external environment.
14.	Component		Represents physical modules which are a collection of components.
15.	Node		Represents physical modules which are a collection of components.
16.	Data Process/State		A circle in DFD represents a state or process which has been triggered due to some event or action.
17.	External entity		Represents external entities such as keyboard, sensors, etc.

18.	Transition		Represents communication that occurs between processes.
19.	Object Lifeline		Represents the vertical dimensions that the object communications.
20.	Message		Represents the message exchanged.

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Netflix is a company that handles a big collection of television programs and movies, by streaming it at any time via online (computers or TV). This firm is profitable because the users do a monthly payment to get access to the platform. However, the clients can cancel their subscriptions at any time (Amatriain, 2013). Therefore, it is vital for the business to keep the users hooked to the platform and not to lose their interest. This is where recommendation systems start to play an important role, it is pivotal to provide valuable suggestions to users (Ricci et al., 2010). The recommendation systems are increasing their popularity among the service providers, because they help to increase the number of items sold, offer a diverse selection of items, the user satisfaction increases, as well as the user fidelity to the company, and they are quite helpful to have a better understanding of what the user wants (Ricci et al., 2010). Then, it is easier to lead the user to make better decisions from a wide variety of cinematographic products.

The recommender systems take into account not only information about the users but also about the items they consume; comparison with other products, and so on and so forth (Hahsler, 2014). Nevertheless, there are many algorithms available to perform a recommendation system. For instance, (i) Popularity, where only the most popular items are recommended (ii) Collaborative Filtering, which looks for patterns in the user activity to produce user-specific recommendations (Breese, Heckerman, and Kadie, 1998); (iii) Content-based Filtering, the recommendation of items with similar information the user has liked or used in the past (description, topic, among others) (Aggarwal, 2016) ; (iv) Hybrid Approaches, combines the two algorithms mentioned above (Adomavicius and Tuzhilin, 2005).

1.2 OBJECTIVE

Then it's easier to get the user to make better decisions from a wide variety of movie products. The idea behind the Netflix recommendation system is to recommend the most popular movies to users. They could be the most-watched, or also the ones with the highest ratings. The popularity of recommendations can be built based on usage data and article content.

1.3 Existing System

In existing system, they are implementing traditional recommendations system to predict the top Netflix movies.

13.1 Existing System Disadvantages

- Less accuracy
- Difficult to predict

.

1.3.2 LITERATURE SURVEY:

Title: Toward the Next Generation of Recommender Systems: A Survey of the State-of-the-Art and Possible Extensions

Year: 2005

Authors: Adomavicius, Gediminas and Alexander Tuzhilin

Description: This paper presents an overview of the field of recommender systems and describes the current generation of recommendation methods that are usually classified into the following three main categories: content-based, collaborative, and hybrid recommendation approaches. This paper also describes various limitations of current recommendation methods and discusses possible extensions that can improve recommendation capabilities and make recommender systems applicable to an even broader range of applications. These extensions include, among others, an improvement of understanding of users and items, incorporation of the contextual information into the recommendation process, support for multi criteria ratings, and a provision of more flexible and less intrusive types of recommendations.

Title : Mining Large Streams of User Data for Personalized Recommendations

Year: 2013

Author: Amatriain, Xavier

Description: The Netflix Prize put the spotlight on the use of data mining and machine learning methods for predicting user preferences. Many lessons came out of the competition. But since then, Recommender Systems have evolved. This evolution has been driven by the greater availability of different kinds of user data in industry and the interest that the area has drawn among the research community. The goal of this paper is to give an up-to-date overview of the use of data mining approaches for personalization and recommendation. Using Netflix personalization as a motivating use case, I will describe the use of different kinds of data and machine learning techniques. After introducing the traditional approaches to recommendation, I highlight some of the main lessons learned from the Netflix Prize. I then describe the use of recommendation and personalization techniques at Netflix. Finally, I pinpoint the most promising current research avenues and unsolved problems that deserve attention in this domain.

Title: Data Mining Methods for Recommender Systems

Year: 2011

Author: Aggarwal, Charu C.

Description: In this chapter, we give an overview of the main Data Mining techniques used in the context of Recommender Systems. We first describe common preprocessing methods such as sampling or dimensionality reduction. Next, we review the most important classification techniques, including Bayesian Networks and Support Vector Machines. We describe the k-means clustering algorithm and discuss several alternatives. We also present association rules and related algorithms for an efficient training process. In addition to introducing these techniques, we survey their uses in Recommender Systems and present cases where they have been successfully applied.

Title: Mining Large Streams of User Data for Personalized Recommendations

Author: Amatriain, Xavier.

Year: 2013

Description: The Netflix Prize put the spotlight on the use of data mining and machine learning methods for predicting user preferences. Many lessons came out of the competition. But since then, Recommender Systems have evolved. This evolution has been driven by the greater availability of different kinds of user data in industry and the interest that the area has drawn among the research community. The goal of this paper is to give an up-to-date overview of the use of data mining approaches for personalization and recommendation. Using Netflix personalization as a motivating use case, I will describe the use of different kinds of data and machine learning techniques. After introducing the traditional approaches to recommendation, I highlight some of the main lessons learned from the Netflix Prize. I then describe the use of recommendation and personalization techniques at Netflix. Finally, I pinpoint the most promising current research avenues and unsolved problems that deserve attention in this domain.

Title: Matrix factorization model in collaborative filtering algorithms: A survey

Author: Bokde, Dheeraj, Sheetal Girase, and Debajyoti Mukhopadhyay

Year: 2015

Description: Recommendation Systems (RSs) are becoming tools of choice to select the online information relevant to a given user. Collaborative Filtering (CF) is the most popular approach to build Recommendation System and has been successfully employed in many applications. Collaborative Filtering algorithms are much explored technique in the field of Data Mining and Information Retrieval. In CF, past user behavior are analyzed in order to establish connections between users and items to recommend an item to a user based on opinions of other users. Those customers, who had similar likings in the past, will have similar likings in the future. In the past decades due to the rapid growth of Internet usage, vast amount of data is generated and it has become a challenge for CF algorithms. So, CF faces issues with sparsity of rating matrix and growing nature of data. These challenges are well taken care of by Matrix Factorization (MF). In this paper we are going to discuss different Matrix Factorization models such as Singular Value Decomposition (SVD), Principal Component Analysis (PCA) and Probabilistic Matrix Factorization (PMF). This paper attempts to present a comprehensive survey of MF model like SVD to address the challenges of CF algorithms, which can be served as a roadmap for research and practice in this area.

1.4 Proposed System

In proposed system, we are implementing Netflix recommendation system by using Machine Learning model termed Pearson R Co-relation.

1.4.1 Proposed System Advantages

- Better accuracy
- Easy to predict

CHAPTER 2

PROJECT DESCRIPTION

2.1 GENERAL:

The idea behind the Netflix recommendation system is to recommend the most popular movies to users. They could be the most-watched, or also the ones with the highest ratings. The popularity of recommendations can be built based on usage data and article content.

2.2 METHODOLOGIES

- 1. MODULE:Data Gathering**
- 2. Data Preparation**
- 3. Data Reduction**
- 4. Recommendation Algorithm**
- 5. Prediction**

MODULE DESCRIPTION

➤ Data Gathering:

To build and develop Machine Learning models, you must first acquire the relevant dataset. This dataset will be comprised of data gathered from multiple and disparate sources which are then combined in a proper format to form a dataset. Dataset formats differ according to use cases. For instance, a business dataset will be entirely different from a medical dataset. In our project the dataset gives the Kaggle and it has a dataset namely movie_titles.csv

➤ Data Preparation:

Data preparation is the process of cleaning and transforming raw data prior to processing and analysis. It is an important step prior to processing and often involves reformatting data, making corrections to data and the combining of data sets to enrich data. The Movie ID column is a mess. We're going to improve this by first creating a NumPy array with the correct length, then adding the entire array as a column in the main data frame.

➤ Data Reduction:

Data reduction is the process of reducing the amount of capacity required to store data. Data reduction can increase storage efficiency and reduce costs. Storage vendors will often describe storage capacity in terms of raw capacity and effective capacity, which refers to data after the reduction. The dataset is super huge. Therefore, we need to reduce the data volume by improving data quality. We can use these two approaches:

- Delete the movie with too many reviews (they are relatively unpopular)
- Remove the customer who gives too less notice (he is relatively less active)

➤ Recommendation Algorithm:

The Pearson correlation coefficient (named for Karl Pearson) can be used to summarize the strength of the linear relationship between two data samples. The Pearson's correlation coefficient is calculated as the covariance of the two variables divided by the product of the

standard deviation of each data sample. Now let's build the Netflix recommendation system by using Pearson correlation algorithm.

➤ **Prediction:**

Prediction refers to the output of an algorithm after it has been trained on a historical dataset and applied to new data when forecasting the likelihood of a particular outcome. In this project, it will predict the top10 movies based on Pearson correlation algorithm.

2.3 TECHNIQUE USED OR ALGORITHM USED

➤ **Recommendation Algorithm:**

The Pearson correlation coefficient (named for Karl Pearson) can be used to summarize the strength of the linear relationship between two data samples. The Pearson's correlation coefficient is calculated as the covariance of the two variables divided by the product of the standard deviation of each data sample. Now let's build the Netflix recommendation system by using Pearson correlation algorithm.

CHAPTER 3

REQUIREMENTS ENGINEERING

3.1 GENERAL

These are the requirements for doing the project. Without using these tools and software's we can't do the project. So we have two requirements to do the project. They are

1. Hardware Requirements.
2. Software Requirements.

3.2 HARDWARE REQUIREMENTS

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system.

They are used by software engineers as the starting point for the system design. It should state what the system does and not how it should be implemented.

- PROCESSOR : DUAL CORE 2 DUOS.
- RAM : 4GB DD RAM
- HARD DISK : 250 GB

3.3 SOFTWARE REQUIREMENTS

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the teams and tracking the team's progress throughout the development activity.

- OPERATING SYSTEM : WINDOWS 7/8/10
- PLATFORM : SPYDER3
- PROGRAMMING LANGUAGE : PYTHON, HTML
- FRONT END : SPYDER3

3.4 FUNCTIONAL REQUIREMENTS

A functional requirement defines a function of a software-system or its component. A function is described as a set of inputs, the behavior, then it's easier to get the user to make better decisions from a wide variety of movie products. The idea behind the Netflix recommendation system is to recommend the most popular movies to users. They could be the most-watched, or also the ones with the highest ratings. The popularity of recommendations can be built based on usage data and article content.

3.5 NON-FUNCTIONAL REQUIREMENTS

The major non-functional Requirements of the system are as follows

➤ **Usability**

The system is designed with completely automated process hence there is no or less user intervention.

➤ **Reliability**

The system is more reliable because of the qualities that are inherited from the chosen platform java. The code built by using java is more reliable.

➤ **Performance**

This system is developing in the high level languages and using the advanced front-end and back-end technologies it will give response to the end user on client system with in very less time.

➤ **Supportability**

The system is designed to be the cross platform supportable. The system is supported on a wide range of hardware and any software platform, which is having JVM, built into the system.

➤ **Implementation**

The system is implemented in web environment using struts framework. The apache tomcat is used as the web server and windows xp professional is used as the platform. Interface the user interface is based on Struts provides HTML Tag

CHAPTER 4

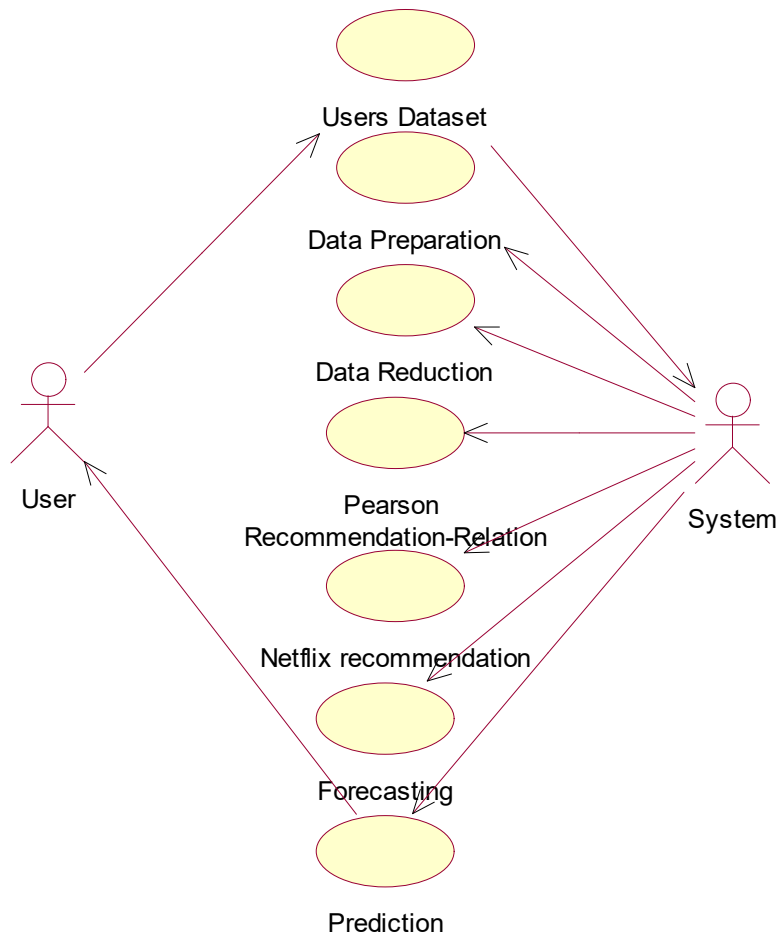
DESIGN ENGINEERING

4.1 GENERAL

Design Engineering deals with the various UML [Unified Modelling language] diagrams for the implementation of project. Design is a meaningful engineering representation of a thing that is to be built. Software design is a process through which the requirements are translated into representation of the software. Design is the place where quality is rendered in software engineering. Design is the means to accurately translate customer requirements into finished product.

UML Diagrams

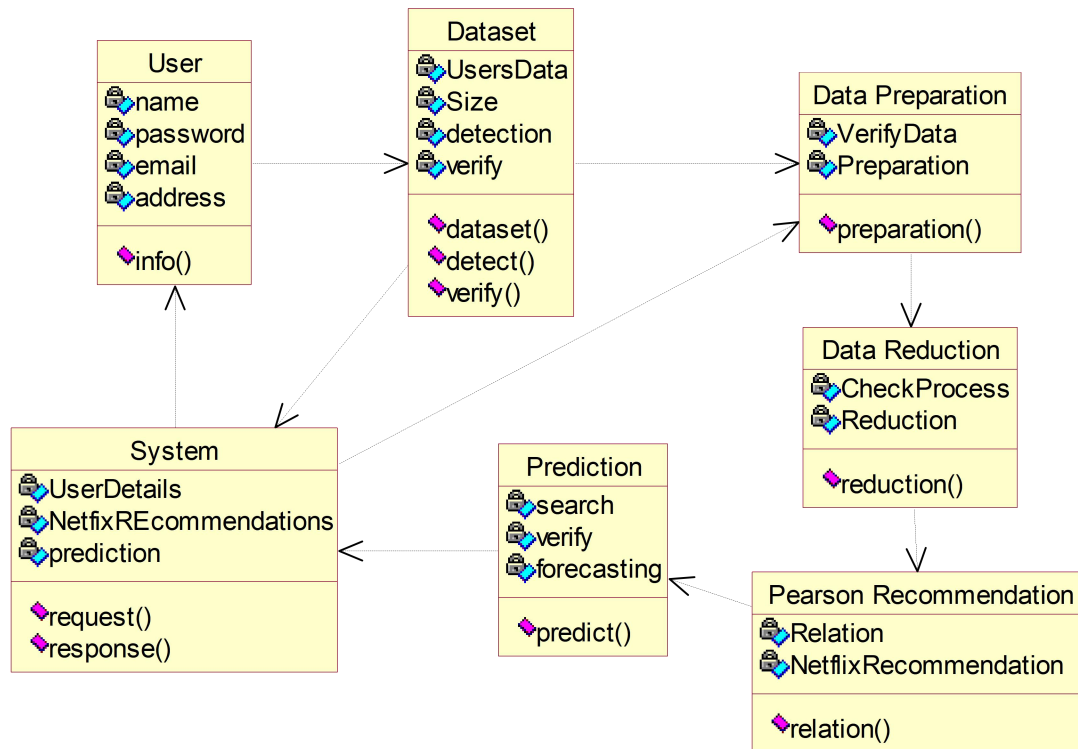
USE CASE DIAGRAM:



EXPLANATION:

The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted. The above diagram consists of user as actor. Each will play a certain role to achieve the concept.

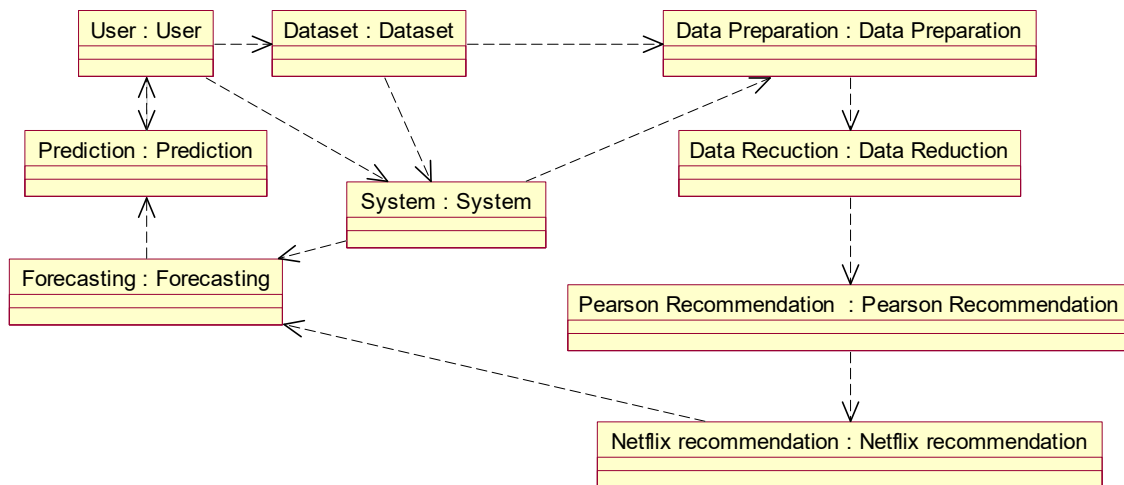
CLASS DIAGRAM:



EXPLANATION:

In this class diagram represents how the classes with attributes and methods are linked together to perform the verification with security. From the above diagram shown the various classes involved in our project.

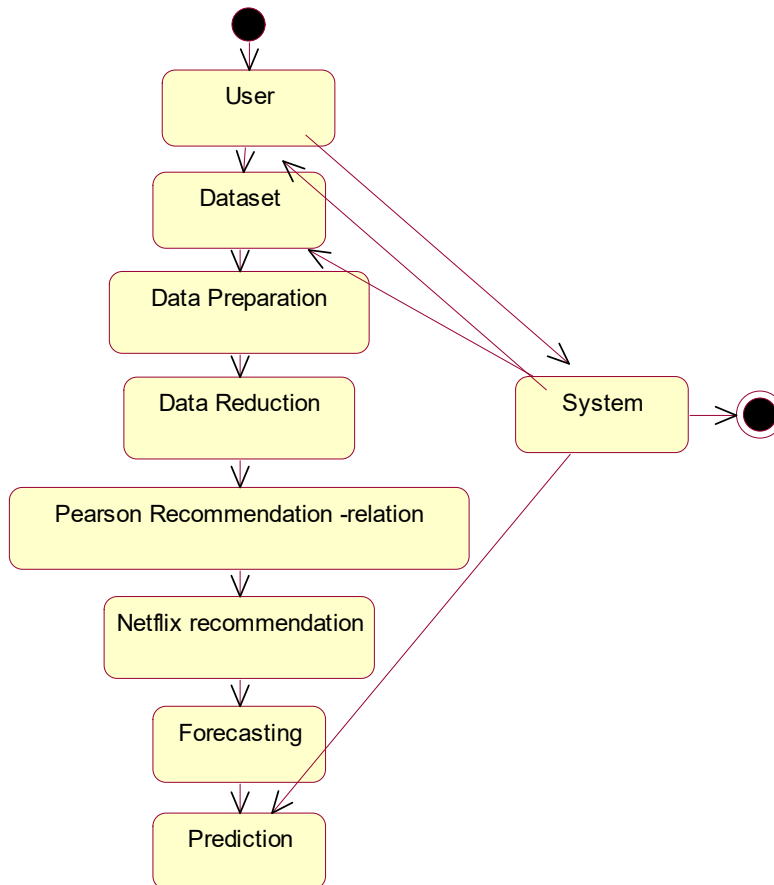
OBJECT DIAGRAM:



EXPLANATION:

In the above digram tells about the flow of objects between the classes. It is a diagram that shows a complete or partial view of the structure of a modeled system. In this object diagram represents how the classes with attributes and methods are linked together to perform the verification with security.

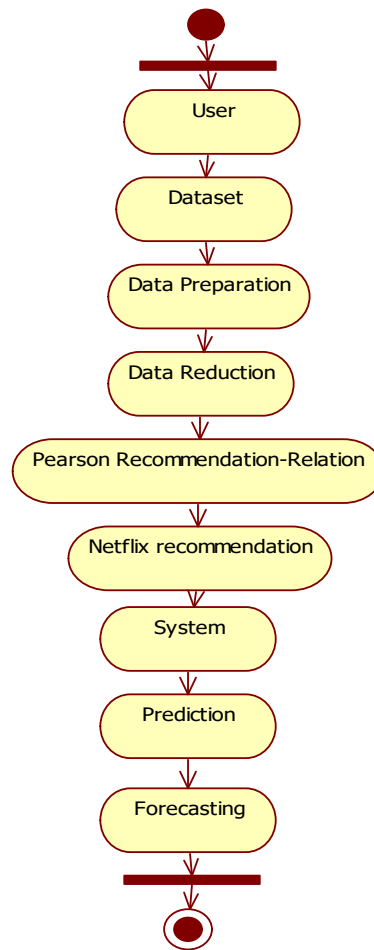
STATE DIAGRAM:



EXPLANATION:

State diagram are a loosely defined diagram to show workflows of stepwise activities and actions, with support for choice, iteration and concurrency. UML, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. UML activity diagrams could potentially model the internal logic of a complex operation. In many ways UML activity diagrams are the object-oriented equivalent of flow charts and data flow diagrams (DFDs) from structural development.

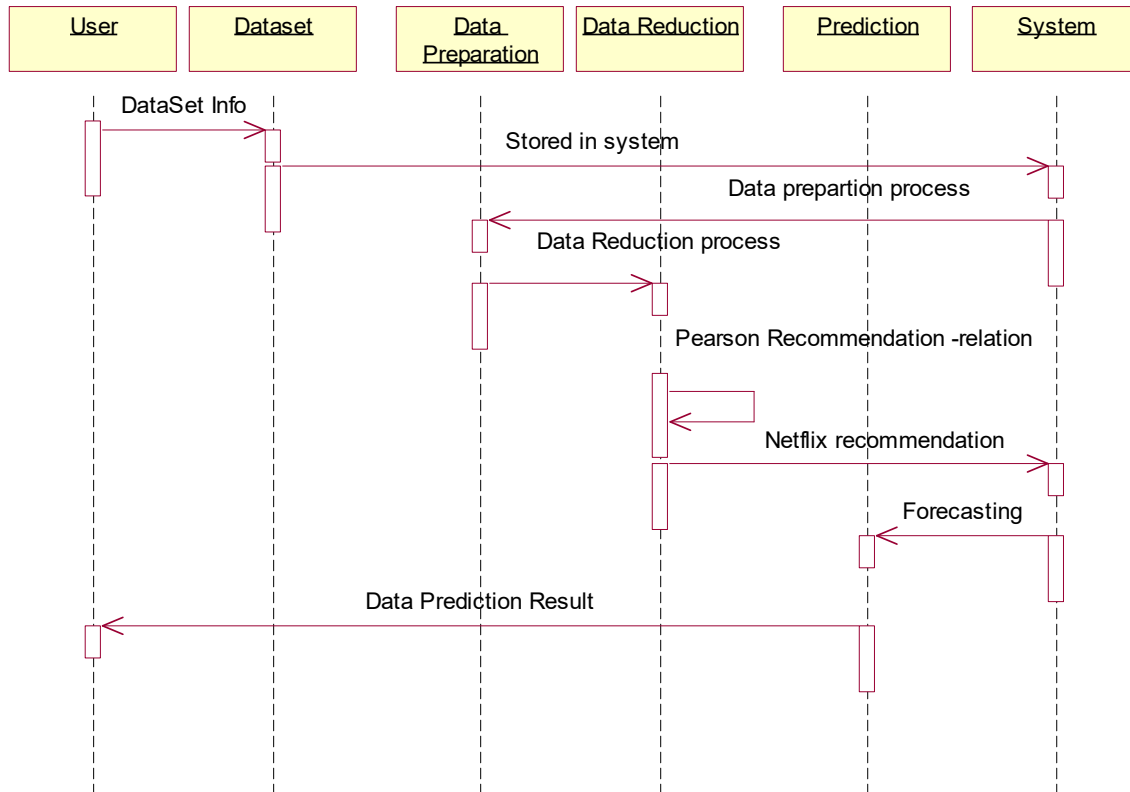
ACTIVITY DIAGRAM:



EXPLANATION:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

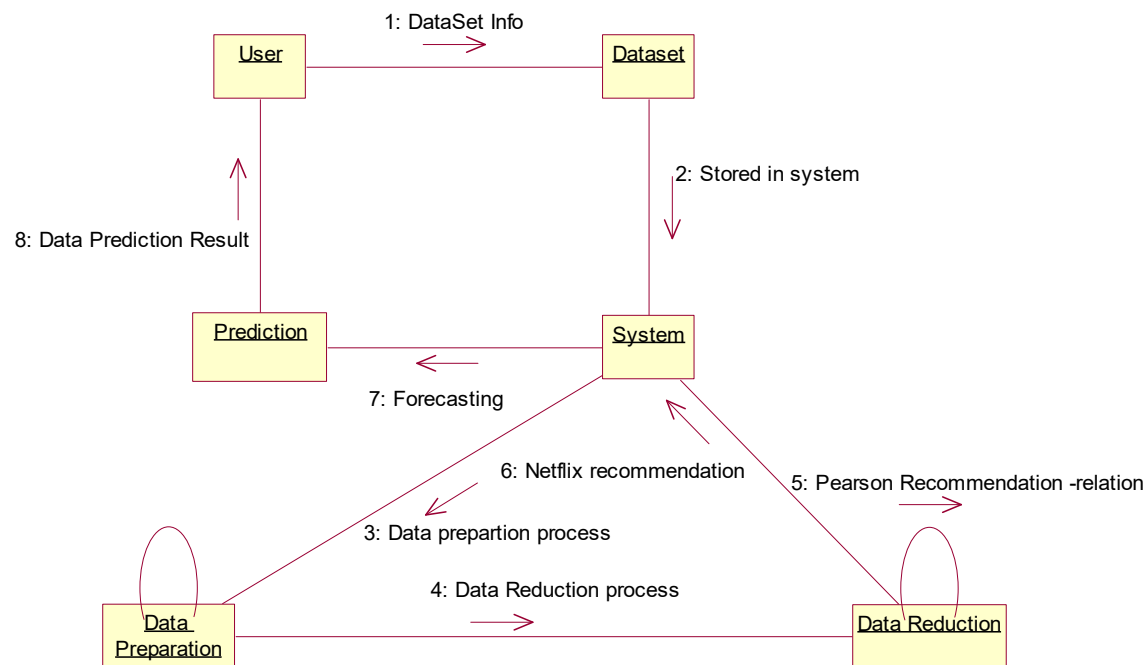
SEQUENCE DIAGRAM:



EXPLANATION:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario.

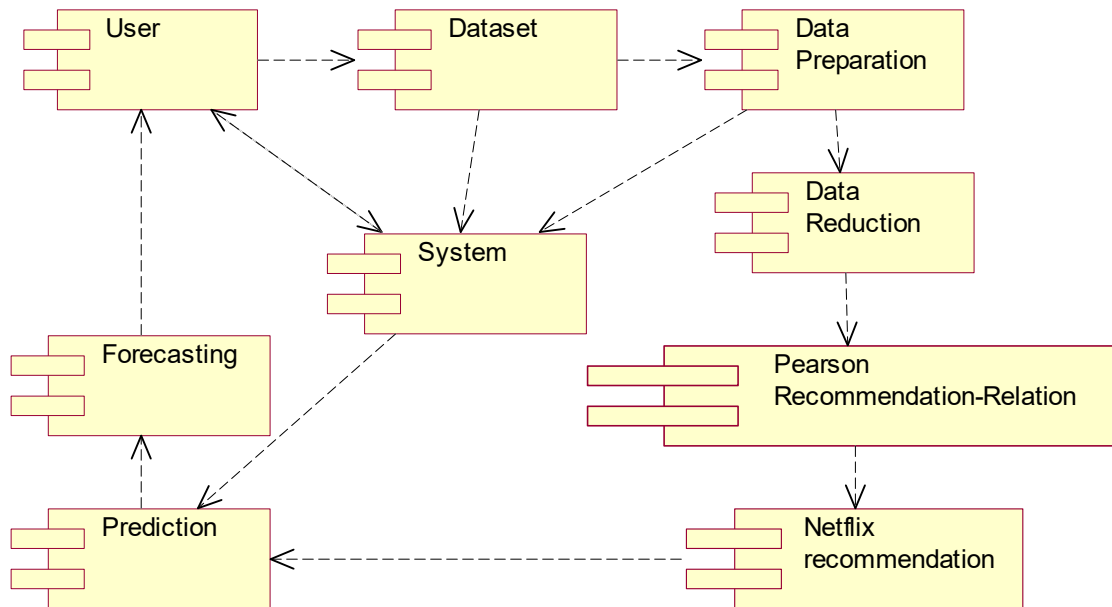
COLLABORATION DIAGRAM:



EXPLANATION:

A collaboration diagram, also called a communication diagram or interaction diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). The concept is more than a decade old although it has been refined as modeling paradigms have evolved.

COMPONENT DIAGRAM:

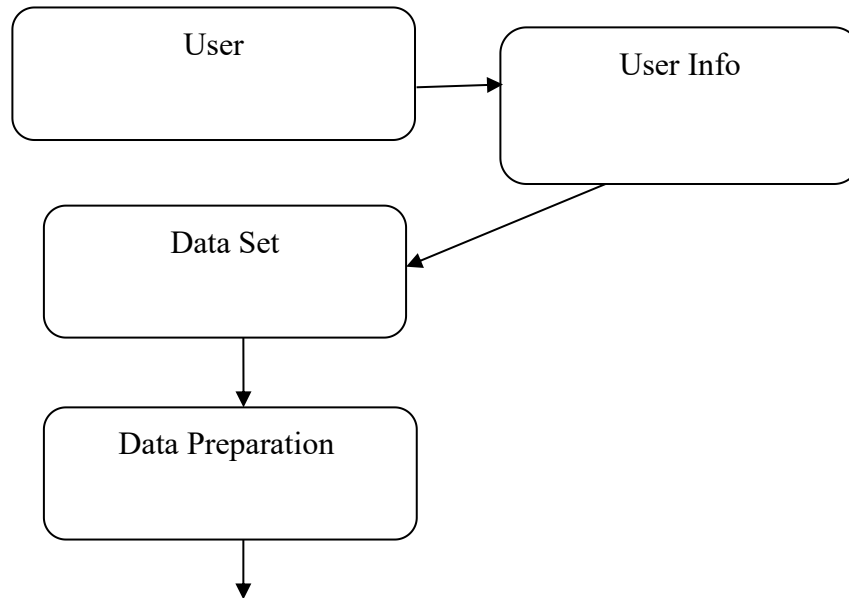


EXPLANATION:

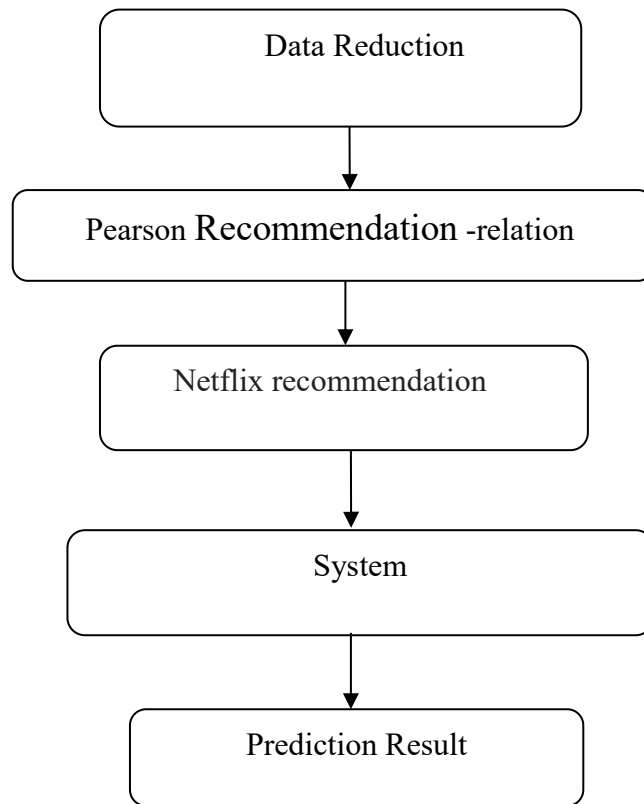
In the Unified Modeling Language, a component diagram depicts how components are wired together to form larger components and or software systems. They are used to illustrate the structure of arbitrarily complex systems. User gives main query and it converted into sub queries and sends through data dissemination to data aggregators. Results are to be showed to user by data aggregators. All boxes are components and arrow indicates dependencies.

DATA FLOW DIAGRAM:

Level-0:



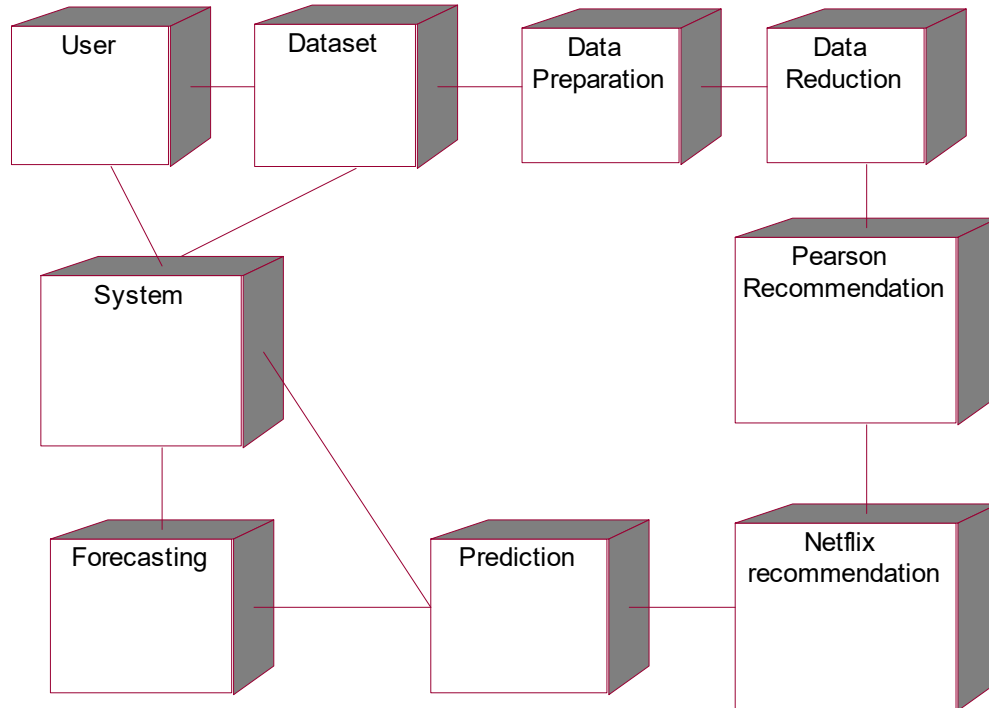
Level-1



EXPLANATION:

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design). A DFD shows what kinds of data will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of processes, or information about whether processes will operate in sequence or in parallel.

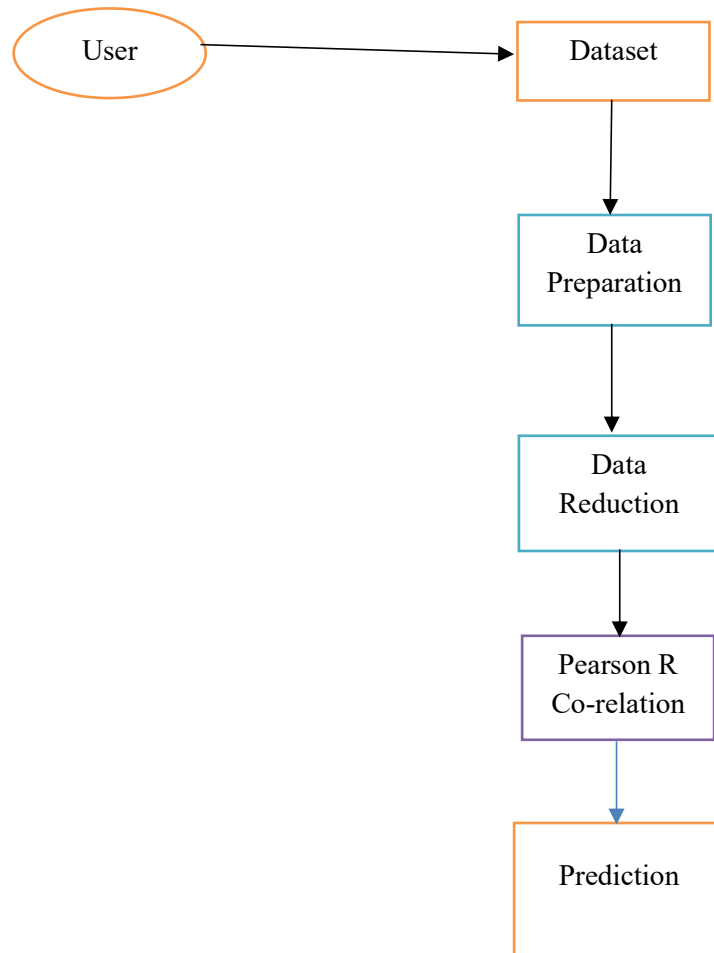
Deployment Diagram



EXPLANATION:

In the Unified Modeling Language, a component diagram depicts how components are wired together to form larger components and or software systems. They are used to illustrate the structure of arbitrarily complex systems. User gives main query and it converted into sub queries and sends through data dissemination to data aggregators. Results are to be showed to user by data aggregators. All boxes are components and arrow indicates dependencies.

SYSTEM ARCHITECTURE:



System Architecture

CHAPTER 5

DEVELOPMENT TOOLS

Python

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

History of Python

Python was developed by Guido van Rossum in the late eighties and early nineties at the National Research Institute for Mathematics and Computer Science in the Netherlands.

Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, SmallTalk, and Unix shell and other scripting languages.

Python is copyrighted. Like Perl, Python source code is now available under the GNU General Public License (GPL).

Python is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress.

Importance of Python

- **Python is Interpreted** – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive** – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented** – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.

- **Python is a Beginner's Language** – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Features of Python

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases** – Python provides interfaces to all major commercial databases.
- **GUI Programming** – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** – Python provides a better structure and support for large programs than shell scripting.

Apart from the above-mentioned features, Python has a big list of good features, few are listed below –

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.
- It provides very high-level dynamic data types and supports dynamic type checking.
- IT supports automatic garbage collection.
- It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.

Libraries used in python:

- numpy - mainly useful for its N-dimensional array objects.
- pandas - Python data analysis library, including structures such as dataframes.
- matplotlib - 2D plotting library producing publication quality figures.
- scikit-learn - the machine learning algorithms used for data analysis and data mining tasks.



Figure : NumPy, Pandas, Matplotlib, Scikit-learn

CHAPTER 6

IMPLEMENTATION

6.1 IMPLEMENTATION

```
import math

from flask import Flask, render_template, request

import pandas as pd

from sklearn.feature_extraction.text import CountVectorizer

from sklearn.metrics.pairwise import cosine_similarity


def prepare_data(x):

    return str.lower(x.replace(" ", ""))
```

```
def create_soup(x):

    return x['Genre'] + ' ' + x['Tags'] + ' ' + x['Actors'] + ' ' + x['ViewerRating']
```

```
def get_recommendations(title, cosine_sim):

    global result

    title=title.replace(' ', '').lower()

    idx = indices[title]

    sim_scores = list(enumerate(cosine_sim[idx]))

    sim_scores = sorted(sim_scores, key=lambda x: x[1], reverse=True)

    sim_scores = sim_scores[1:51]

    movie_indices = [i[0] for i in sim_scores]

    result = netflix_data.iloc[movie_indices]

    result.reset_index(inplace = True)

    return result
```

```

netflix_data = pd.read_csv('NetflixDataset.csv',encoding='latin-1', index_col = 'Title')

netflix_data.index = netflix_data.index.str.title()

netflix_data = netflix_data[~netflix_data.index.duplicated()]

netflix_data.rename(columns={'View Rating':'ViewerRating'}, inplace=True)

Language = netflix_data.Languages.str.get_dummies(',')

Lang = Language.columns.str.strip().values.tolist()

Lang = set(Lang)

Titles = set(netflix_data.index.to_list())

```

```

netflix_data['Genre'] = netflix_data['Genre'].astype('str')

netflix_data['Tags'] = netflix_data['Tags'].astype('str')

netflix_data['IMDb Score'] = netflix_data['IMDb Score'].apply(lambda x: 6.6 if math.isnan(x) else x)

netflix_data['Actors'] = netflix_data['Actors'].astype('str')

netflix_data['ViewerRating'] = netflix_data['ViewerRating'].astype('str')

new_features = ['Genre', 'Tags', 'Actors', 'ViewerRating']

selected_data = netflix_data[new_features]

for new_feature in new_features:

    selected_data.loc[:, new_feature] = selected_data.loc[:, new_feature].apply(prepare_data)

selected_data.index = selected_data.index.str.lower()

selected_data.index = selected_data.index.str.replace(" ", '')

selected_data['soup'] = selected_data.apply(create_soup, axis = 1)

```

```

count = CountVectorizer(stop_words='english')

count_matrix = count.fit_transform(selected_data['soup'])

cosine_sim2 = cosine_similarity(count_matrix, count_matrix)

selected_data.reset_index(inplace = True)

indices = pd.Series(selected_data.index, index=selected_data['Title'])

result = 0

df = pd.DataFrame()

```

```
app = Flask(__name__)
```

```
@app.route('/')
```

```
def index():
```

```
    return render_template('index.html', languages = Lang, titles = Titles)
```

```
@app.route('/about',methods=['POST'])
```

```
def getvalue():
```

```
    global df
```

```
    movienames = request.form.getlist('titles')
```

```
    languages = request.form.getlist('languages')
```

```
    for moviename in movienames:
```

```
        get_recommendations(moviename,cosine_sim2)
```

```
        for language in languages:
```

```
            df = pd.concat([result[result['Languages'].str.count(language) > 0], df], ignore_index=True)
```

```
df.drop_duplicates(keep = 'first', inplace = True)
```

```
df.sort_values(by = 'IMDb Score', ascending = False, inplace = True)
```

```
images = df['Image'].tolist()
```

```
titles = df['Title'].tolist()
```

```
@app.route('/moviepage/<name>')
```

```
def movie_details(name):
```

```
    global df
```

```
    details_list = df[df['Title'] == name].to_numpy().tolist()
```

```
    return render_template('moviepage.html', details = details_list[0])
```

```
return render_template('result.html', titles = titles, images = images)
```

```
if __name__ == '__main__':
```

```
    app.run(debug=False)
```

CHAPTER 7

GENERAL

This project implements like application using python and the Server process is maintained using the SOCKET & SERVERSOCKET and the Design part is played by Cascading Style Sheet.

SNAPSHOTS



CHAPTER 8

SOFTWARE TESTING

8.1 GENERAL

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

8.2 DEVELOPING METHODOLOGIES

The test process is initiated by developing a comprehensive plan to test the general functionality and special features on a variety of platform combinations. Strict quality control procedures are used. The process verifies that the application meets the requirements specified in the system requirements document and is bug free. The following are the considerations used to develop the framework from developing the testing methodologies.

8.3 Types of Tests

8.3.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program input produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

8.3.2 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

8.3.3 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

8.3.4 Performance Test

The Performance test ensures that the output be produced within the time limits, and the time taken by the system for compiling, giving response to the users and request being send to the system for to retrieve the results.

8.3.5 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

8.3.6 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Acceptance testing for Data Synchronization:

- The Acknowledgements will be received by the Sender Node after the Packets are received by the Destination Node
- The Route add operation is done only when there is a Route request in need
- The Status of Nodes information is done automatically in the Cache Updation process

8.2.7 Build the test plan

Any project can be divided into units that can be further performed for detailed processing. Then a testing strategy for each of this unit is carried out. Unit testing helps to identify the possible bugs in the individual component, so the component that has bugs can be identified and can be rectified from errors.

CHAPTER 9

FUTURE ENHANCEMENT

9.1 FUTURE ENHANCEMENT

It is part of our future work. Moreover, we believe that the face reconstruction part of our method can be further improved by combining “Pixel wise” and “Identity-Preserving” approaches into one optimization objective. We are also planning to explore this path in our further studies.

CHAPTER 10

CONCLUSION & REFERENCE

10.1 CONCLUSION

In this work, we have proposed a new effective method for synthetic aging of human faces based on Age Conditional Generative Adversarial Network (Age-cGAN). The method is composed of two steps: (1) input face reconstruction requiring the solution of an optimization problem in order to find an optimal latent approximation z^* , (2) and face aging it-self performed by a simple change of condition y at the input of the generator. The cornerstone of our method is the novel “Identity-Preserving” latent vector optimization approach allowing to preserve the original person’s identity in the reconstruction. This approach is universal meaning that it can be used to preserve identity not only for face aging but also for other face alterations (e.g. adding a beard, sunglasses etc.)

Our face aging method can be used for synthetic augmentation of face datasets and for improving the robustness of face recognition solutions in cross-age scenarios.

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